

UNIT-5 | Lecture 6

Query Optimization Algorithms — Cost-Based Optimization

Coverage: Hour 6

Learning Objectives

Understand cost-based query optimization.

Understand statistics, execution plans, and plan selection.

Review the complete optimization workflow.

Cost-Based Optimization

A **cost-based optimizer (CBO)** generates alternative execution plans, estimates their costs, and chooses a plan with low estimated cost.

Cost estimates can include disk I/O, CPU, memory, and intermediate-result sizes.

Statistics

Optimizers use statistics such as **number of rows, number of pages, value distributions, distinct values, and index information**.

Accurate statistics improve cardinality and cost estimates.

Execution Plan

An execution plan specifies how the DBMS will execute a query, including access methods, join order, join algorithms, sorting, grouping, and other operations.

The optimizer compares alternative plans and selects one based on estimated cost.

Complete Optimization Workflow

SQL Query → Parse/Translate → Relational Expression → Generate Equivalent Plans → Estimate Costs → Select Best Plan → Execute

Heuristic vs Cost-Based

Heuristic	Cost-Based
Uses transformation rules	Uses estimated execution costs
Usually faster optimization	May spend more effort comparing plans
Good general rules	Can adapt to data statistics and available access paths
May not find the cheapest plan	Attempts to select a low-cost plan

Key Takeaway

Query optimization transforms a query into efficient execution plans; cost-based optimization uses statistics and cost estimates to select among alternatives.